

- 34** A straight copper wire, of diameter $4.2 \times 10^{-4} \text{ m}$, has a number density of free electrons of $8.5 \times 10^{28} \text{ m}^{-3}$. In a given time interval, a charge of 0.15 C moves through the wire.

What is the average displacement of the free electrons along the wire in this time interval?

- A** $3.3 \times 10^{-8} \text{ m}$ **B** $2.0 \times 10^{-5} \text{ m}$ **C** $8.0 \times 10^{-5} \text{ m}$ **D** $2.5 \times 10^{-4} \text{ m}$